

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Coursework:** Data Structures & Algorithms, OOP, Computer Architecture, Linear Algebra, Discrete Math

EXPERIENCE

Secure and Efficient Autonomous Systems Lab

Undergraduate Research Assistant

May 2026 – Present

East Lansing, MI

- Built the verifier as a C++ ROS2 node running per-step geometric and semantic checks on each motion plan, the gate the decomposition attack is built to defeat.
- Architected the node graph to run byte-identical in Gazebo and on a physical RoboMaster EP Core, isolating the sim-to-hardware swap to the URDF and controller layer.
- Reproduced a published adversarial-robotics attack loop (attacker, target, judge, syntax-checker) as the baseline, so the decomposition attack's success rate reads against a credible reference.

Robostangs (FRC Team 548)

Software Captain

Jun 2023 – May 2024

Northville, MI

- Led the software and electrical teams (5 and 6 engineers) for a competition robot that reached the FIRST World Championship and placed top 16 in its division in Houston, the program's strongest world-stage finish since 2014.
- Anchored auto-aim on real-time, vision-corrected localization that fused two AprilTag cameras with swerve odometry on a dedicated 250Hz thread with latency compensation, holding pose accurate during aggressive motion.

MindMosaic

Software Engineering Intern

Aug 2025 – May 2026

East Lansing, MI

- Migrated hot read paths off recursive SQL self-joins onto a Neo4j graph layer so relationships stay first-class, cutting query latency 48% (850ms to 440ms) and making multi-hop traversals 3x faster than the equivalent SQL.
- Authored an AWS deployment script and managed the environment as code with Terraform, turning a manual multi-service bring-up into a repeatable, version-controlled provision so the polyglot stack deploys identically every time.

PROJECTS

Dadei | *Python, FastAPI, Redis, Docker, PostgreSQL*

dadei.app

Ambient voice assistant that turns overheard conversation into human-approved calendar, email, and task actions

- Decoupled background workers from the live API behind a Redis Streams event bus with consumer groups, dead-letter routing after five failed deliveries, and stale-claim recovery, giving at-least-once delivery across process boundaries that Redis pub/sub silently dropped on disconnect.
- Implemented delayed action auto-fire on wall-clock-relative job scheduling with a one-second flush backstop, keeping countdown auto-fire correct across worker timezones and tight to the user-visible timer instead of drifting to a 20-second heartbeat.

fliks | *Go, Docker*

github.com/fliks-gg

Multi-game platform that crowd-validates gameplay skill and mints shareable certification cards

- Chose a single Go monolith over a microservice split, keeping auth, rate-limiting, and rating aggregation in-process because premature decomposition is the harder call to defend at pre-scale, one deploy unit against one datastore.
- Built a self-hosted video transcoding pipeline with a concurrent Go worker that pulls upload jobs, transcodes, and writes processed video back to object storage, owning the media path end to end instead of renting a managed service.

TECHNICAL SKILLS

Languages C++, Python, Java, Go, JavaScript/TypeScript, SQL, Bash

Frameworks FastAPI

Cloud & Infrastructure AWS (EC2, S3, RDS), Docker, Terraform

Robotics ROS2, Gazebo, Linux, Git, AprilTag

Databases PostgreSQL, Neo4j, Redis